

IF2240 - Basis Data

Praktikum 0

Eksplorasi SQL

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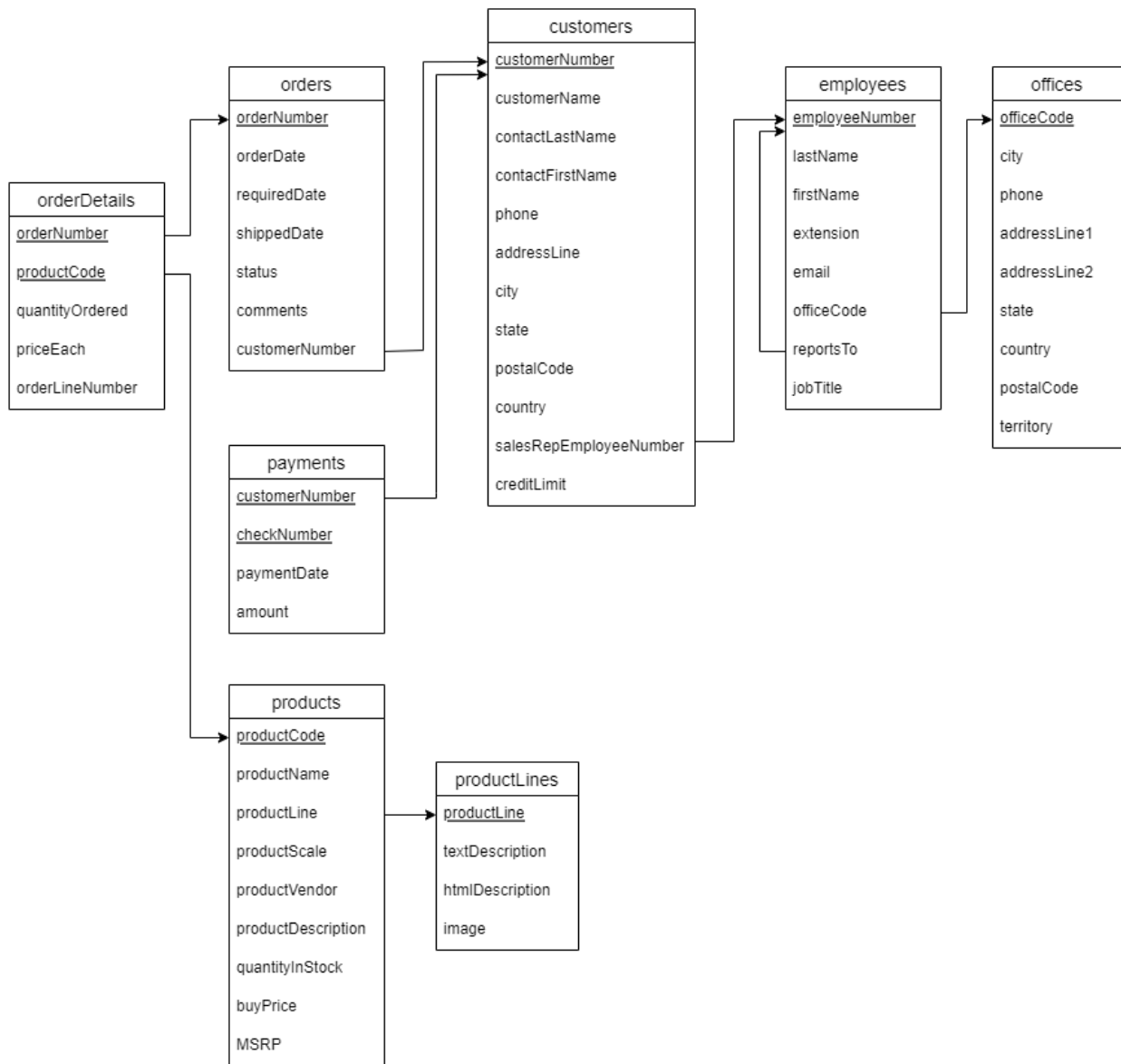


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Sekolah Teknik Elektro dan Informatika
Institut Teknologi Bandung
2026**

Multinational Company



Legenda

No	Nama Tabel	Keterangan
1	customers	Pelanggan dari perusahaan
2	employees	Pegawai dari perusahaan
3	offices	Kantor cabang perusahaan
4	orders	Pemesanan
5	orderDetails	Detail barang yang dipesan untuk pemesanan tertentu
6	payments	Pembayaran yang dilakukan pelanggan
7	products	Produk yang dijual perusahaan
8	productLines	Lini produk

Petunjuk

1. Praktikum 0 bersifat *take home*.
2. Praktikum ini bersifat individual. Segala plagiarisme akan mengakibatkan nilai 0 pada praktikum ini.

I. Soal

1. Buatlah *query* untuk menampilkan seluruh tabel yang terdapat dalam database prak1!

QUERY: SHOW TABLES;

SCREENSHOT:

```
MariaDB [(none)]> USE prak0;  
Database changed  
MariaDB [prak0]> SHOW TABLES;  
+-----+  
| Tables_in_prak0 |  
+-----+  
| customers        |  
| employees        |  
| offices          |  
| orderdetails     |  
| orders           |  
| payments         |  
| productlines     |  
| products         |  
+-----+  
8 rows in set (0.001 sec)  
  
MariaDB [prak0]> |
```

2. Buatlah *query* untuk menampilkan detail informasi atribut dari entitas customers yang terdapat dalam database prak0!

QUERY: DESCRIBE customers;

SCREENSHOT:

```
MariaDB [prak0]> DESCRIBE customers;
```

Field	Type	Null	Key	Default	Extra
customerNumber	int(11)	NO	PRI	NULL	
customerName	varchar(50)	NO		NULL	
contactLastName	varchar(50)	NO		NULL	
contactFirstName	varchar(50)	NO		NULL	
phone	varchar(50)	NO		NULL	
addressLine1	varchar(50)	NO		NULL	
addressLine2	varchar(50)	YES		NULL	
city	varchar(50)	NO		NULL	
state	varchar(50)	YES		NULL	
postalCode	varchar(15)	YES		NULL	
country	varchar(50)	NO		NULL	
salesRepEmployeeNumber	int(11)	YES	MUL	NULL	
creditLimit	decimal(10,2)	YES		NULL	

13 rows in set (0.082 sec)

3. Buatlah *query* untuk menampilkan seluruh nama pengunjung dalam huruf kecil!

HINT: gunakan LOWER()

QUERY: SELECT LOWER(customerName) FROM customers;

SCREENSHOT:

```
MariaDB [prak0]> SELECT LOWER(customerName) FROM customers;
```

LOWER(customerName)
atelier graphique
signal gift stores
australian collectors, co.
la rochelle gifts
baane mini imports
mini gifts distributors ltd.
havel & zbyszek co
blauer see auto, co.
mini wheels co.
land of toys inc.
euro+ shopping channel
volvo model replicas, co
danish wholesale imports
saveley & henriot, co.
dragon souvenirs, ltd.
muscle machine inc
diecast classics inc.
technics stores inc.
handji gifts& co
herkku gifts
american souvenirs inc
porto imports co.
daedalus designs imports
la corne d'abondance, co.
cambridge collectables co.
gift depot inc.
osaka souvenirs co.
vitachrome inc.
toys of finland, co.
av stores, co.
clover collections, co.
auto-moto classics inc.
uk collectables, ltd.

canadian gift exchange network
online mini collectables
toys4grownups.com
asian shopping network, co
mini caravy
king kong collectables, co.
enaco distributors
boards & toys co.
nat??rlich autos
heintze collectables
qu??bec home shopping network
ang resellers
collectable mini designs co.
giftsbymail.co.uk
alpha cognac
messner shopping network
amica models & co.
lyon souvenirs
auto associ??s & cie.
toms spezialit?ñten, ltd
royal canadian collectables, ltd.
franken gifts, co
anna's decorations, ltd
rovelli gifts
souvenirs and things co.
marta's replicas co.
bg&e collectables
vida sport, ltd
norway gifts by mail, co.
schuyler imports
der hund imports
oulu toy supplies, inc.
petit auto
mini classics
mini creations ltd.
corporate gift ideas co.
down under souvenirs, inc
stylish desk decors, co.

tekni collectables inc. australian gift network, co suominen souvenirs cramer spezialit?nten, ltd classic gift ideas, inc caf imports men 'r' us retailers, ltd. asian treasures, inc. marseille mini autos reims collectables sar distributors, co giftsforhim.com kommission auto gifts4allages.com online diecast creations co. lisboa souvenirs, inc precious collectables collectables for less inc. royale belge salzburg collectables cruz & sons co. l'ordine souvenirs tokyo collectables, ltd auto canal+ petit stuttgart collectable exchange extreme desk decorations, ltd bavarian collectables imports, co. classic legends inc. feuer online stores, inc gift ideas corp. scandinavian gift ideas the sharp gifts warehouse mini auto werke super scale inc. microscale inc. corrida auto replicas, ltd warburg exchange fungiftideas.com	
---	--


```

| anton designs, ltd.
| australian collectables, ltd
| frau da collezione
| west coast collectables co.
| mit vergn??gen & co.
| kremlin collectables, co.
| raanan stores, inc
| iberia gift imports, corp.
| motor mint distributors inc.
| signal collectibles ltd.
| double decker gift stores, ltd
| diecast collectables
| kelly's gift shop
+-----+
122 rows in set (0.047 sec)

```

4. Buatlah *query* untuk menampilkan 5 nama pengunjung yang berasal dari kota Bern!

HINT: gunakan LIMIT untuk membatasi jumlah data yang ditampilkan

QUERY: SELECT customerName FROM customers WHERE city = 'Bern' LIMIT 5;

SCREENSHOT:

```

MariaDB [prak0]> SELECT customerName FROM customers WHERE city = 'Bern' LIMIT 5;
+-----+
| customerName |
+-----+
| Precious Collectables |
+-----+
1 row in set (0.046 sec)

```

5. Buatlah *query* untuk menampilkan nama pegawai yang bekerja di kantor yang berada di negara Australia!

HINT: gunakan CONCAT() untuk menggabungkan lebih dari 1 string

QUERY:

```

SELECT CONCAT(firstName, ' ', lastName)
AS employeesName
FROM employees
WHERE officeCode IN(SELECT officeCode FROM offices WHERE country =
'Australia');

```

Tanpa Subquery:

```

SELECT CONCAT(firstName, ' ', lastName)
AS employeesName
FROM employees AS e, offices AS o
WHERE e.officeCode = o.officeCode AND o.country = 'Australia';

```

SCREENSHOT:

```
MariaDB [prak0]> SELECT CONCAT(firstName, ' ', lastName) AS employeesName FROM employees WHERE officeCode IN(
-> SELECT officeCode FROM offices WHERE country = 'Australia');
+-----+
| employeesName |
+-----+
| William Patterson |
| Andy Fixter      |
| Peter Marsh      |
| Tom King         |
+-----+
4 rows in set (0.001 sec)
```

6. Buatlah *query* yang menampilkan nama pengunjung yang pernah membuat pembayaran dengan jumlah diatas 50000!

QUERY:

```
SELECT customerName
FROM customers
WHERE customerNumber
IN(
  SELECT customerNumber
  FROM payments
  WHERE amount > 50000
);
```

SCREENSHOT:

```

MariaDB [prak0]> SELECT customerName FROM customers WHERE customerNumber IN
-> (SELECT customerNumber FROM payments WHERE amount > 50000);
+-----+
| customerName |
+-----+
| Australian Collectors, Co. |
| Baane Mini Imports |
| Mini Gifts Distributors Ltd. |
| Land of Toys Inc. |
| Euro+ Shopping Channel |
| Danish Wholesale Imports |
| Dragon Souvenirs, Ltd. |
| Muscle Machine Inc |
| Diecast Classics Inc. |
| Technics Stores Inc. |
| Herkku Gifts |
| La Corne D'abondance, Co. |
| AV Stores, Co. |
| Online Mini Collectables |
| Toys4GrownUps.com |
| Heintze Collectables |
| Collectable Mini Designs Co. |
| Auto Associ??s & Cie. |
| Toms Spezialit?nten, Ltd |
| Rovelli Gifts |
| Vida Sport, Ltd |
| Mini Creations Ltd. |
| Corporate Gift Ideas Co. |
| Down Under Souvenirs, Inc |
| Marseille Mini Autos |
| Online Diecast Creations Co. |
| Cruz & Sons Co. |
| L'ordine Souvenirs |
| The Sharp Gifts Warehouse |
| Corrida Auto Replicas, Ltd |
| Diecast Collectables |
| Kelly's Gift Shop |
+-----+
32 rows in set (0.007 sec)

```

7. Buatlah *query* untuk menampilkan nama pengunjung yang dilayani oleh pegawai yang bekerja di kantor yang berada di negara Japan!

QUERY:

```

SELECT DISTINCT c.customerName
FROM customers c
JOIN employees e ON c.salesRepEmployeeNumber = e.employeeNumber
JOIN offices o ON e.officeCode = o.officeCode
WHERE o.country = 'Japan';

```

Tanpa join:

```

SELECT DISTINCT c.customerName
FROM customers c, employees e, offices o

```

```
WHERE c.salesRepEmployeeNumber = e.employeeNumber  
AND e.officeCode = o.officeCode  
AND o.country = 'Japan';
```

SCREENSHOT:

```
MariaDB [prak0]> SELECT DISTINCT c.customerName  
-> FROM customers c  
-> JOIN employees e ON c.salesRepEmployeeNumber = e.employeeNumber  
-> JOIN offices o ON e.officeCode = o.officeCode  
-> WHERE o.country = 'Japan';  
  
+-----+  
| customerName |  
+-----+  
| Dragon Souvenirs, Ltd. |  
| Osaka Souvenirs Co. |  
| King Kong Collectables, Co. |  
| Cruz & Sons Co. |  
| Tokyo Collectables, Ltd |  
+-----+  
5 rows in set (0.004 sec)
```

8. Buatlah *query* untuk menampilkan nama produk yang memiliki harga beli di atas 50 dan pernah dipesan di antara bulan April sampai September (inklusif)!

QUERY:

```
SELECT DISTINCT p.productName  
FROM products p  
JOIN orderDetails od ON p.productCode = od.productCode  
JOIN orders o ON od.orderNumber = o.orderNumber  
WHERE p.buyPrice > 50 AND MONTH(o.orderDate) BETWEEN 4 AND 9;
```

Tanpa Join:

```
SELECT DISTINCT p.productName  
FROM products p, orderDetails od, orders o  
WHERE p.productCode = od.productCode  
AND od.orderNumber = o.orderNumber  
AND p.buyPrice > 50  
AND MONTH(o.orderDate) BETWEEN 4 AND 9;
```

SCREENSHOT:

```

MariaDB [prak0]> SELECT DISTINCT p.productName
-> FROM products p
-> JOIN orderDetails od ON p.productCode = od.productCode
-> JOIN orders o ON od.orderNumber = o.orderNumber
-> WHERE p.buyPrice > 50
-> AND MONTH(o.orderDate) BETWEEN 4 AND 9;

```

productName
1952 Alpine Renault 1300
1996 Moto Guzzi 1100i
2003 Harley-Davidson Eagle Drag Bike
1972 Alfa Romeo GTA
1962 LanciaA Delta 16V
1968 Ford Mustang
2001 Ferrari Enzo
1958 Setra Bus
2002 Suzuki XREO
1969 Corvair Monza
1968 Dodge Charger
1969 Ford Falcon
1957 Chevy Pickup
1969 Dodge Charger
1940 Ford Pickup Truck
1993 Mazda RX-7
1937 Lincoln Berline
1965 Aston Martin DB5
1980s Black Hawk Helicopter
1917 Grand Touring Sedan
1948 Porsche 356-A Roadster
1995 Honda Civic
1998 Chrysler Plymouth Prowler
1964 Mercedes Tour Bus
1932 Model A Ford J-Coupe
1928 Mercedes-Benz SSK
1999 Indy 500 Monte Carlo SS
1913 Ford Model T Speedster
1999 Yamaha Speed Boat
18th Century Vintage Horse Carriage
1903 Ford Model A
1992 Ferrari 360 Spider red

```

| Collectable Wooden Train
| 1917 Maxwell Touring Car
| 1976 Ford Gran Torino
| 1948 Porsche Type 356 Roadster
| 1941 Chevrolet Special Deluxe Cabriolet
| 1970 Triumph Spitfire
| 1904 Buick Runabout
| 1940s Ford truck
| 1957 Corvette Convertible
| 1997 BMW R 1100 S
| 1928 British Royal Navy Airplane
| 18th century schooner
| 1962 Volkswagen Microbus
| 1952 Citroen-15CV
| 1969 Chevrolet Camaro Z28
| 2002 Chevy Corvette
| 1956 Porsche 356A Coupe
| 1992 Porsche Cayenne Turbo Silver
| 1936 Chrysler Airflow
| 1980?ÇÖs GM Manhattan Express
| 1997 BMW F650 ST
| 1974 Ducati 350 Mk3 Desmo
| Diamond T620 Semi-Skirted Tanker
| American Airlines: B767-300
| America West Airlines B757-200
| ATA: B757-300
| F/A 18 Hornet 1/72
| The Titanic
| The Queen Mary

```

```

+-----+
61 rows in set (0.046 sec)

```

9. Buatlah query untuk menampilkan nama lengkap karyawan yang melapor kepada Anthony Bow dan pernah melayani pengunjung yang berasal dari Brickhaven atau Las Vegas!

QUERY:

```

SELECT DISTINCT CONCAT(e.firstName, ' ', e.lastName)
AS employeesName
FROM employees e
JOIN employees m ON e.reportsTo = m.employeeNumber
JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
WHERE m.firstName = 'Anthony'

```

```
AND m.lastName = 'Bow'  
AND c.city IN('Brickhaven', 'Las Vegas');
```

Tanpa Join:

```
SELECT DISTINCT CONCAT(e.firstName, ' ', e.lastName)  
AS employeesName  
FROM employees e, employees m, customers c  
WHERE e.reportsTo = m.employeeNumber  
AND e.employeeNumber = c.salesRepEmployeeNumber  
AND m.firstName = 'Anthony' AND m.lastName = 'Bow'  
AND (c.city='Brickhaven' OR c.city='Las Vegas');
```

SCREENSHOT:

```
MariaDB [prak0]> SELECT DISTINCT CONCAT(e.firstName, ' ', e.lastName) AS employeesName  
-> FROM employees e  
-> JOIN employees m ON e.reportsTo = m.employeeNumber  
-> JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber  
-> WHERE m.firstName = 'Anthony' AND m.lastName = 'Bow' AND c.city IN('Brickhaven', 'Las Vegas');  
  
+-----+  
| employeesName |  
+-----+  
| Leslie Thompson |  
| Julie Firrelli |  
| Steve Patterson |  
+-----+  
3 rows in set (0.001 sec)
```